



Data Structures Quiz

To pass the quiz, you must correctly answer at least 18 of the 20 questions below.

1. What does Big O notation describe in algorithm analysis?

The percentage of code lines executed during a run.

The exact runtime in seconds for a specific computer.

How readable the code is to other developers.

How the time or space grows relative to input size (an upper bound).

✓ Correct!

2. When starting an algorithmic challenge, what is the best first step?

Begin coding immediately to gain momentum.

Write unit tests only after finishing the solution.

Optimize for performance before you understand the problem.

Clarify the problem and constraints with examples and edge cases.

✓ Correct!

3. What is the key difference between dynamic arrays and static arrays?

Dynamic arrays can grow or shrink by resizing; static arrays have a fixed size.

✓ Correct!

Dynamic arrays are faster than static arrays for every operation.



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Static arrays allow duplicate values; dynamic arrays do not.

4. What is the amortized time complexity of appending an element to the end of a dynamic array?

 amortized.

✓ Correct!

5. Why does accessing the k-th element by index in a singly linked list take $O(n)$ time?

Nodes are stored contiguously, so shifting is required.

You must traverse from the head node to the k-th node one by one.

✓ Correct!

The list must be resized before any access.

The index is hashed and looked up in a table.

6. Which feature does a doubly linked list have that a singly linked list does not?

Random access to any index in $O(1)$ time.

Pointers to both next and previous nodes enabling backward traversal.

✓ Correct!

A built-in array buffer for faster iteration.



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7. Which of the following best describes a stack?

First In, First Out (FIFO) with removals at the front.

A circular buffer with constant-time random access.

Last In, First Out (LIFO) with push and pop at the top.

✓ Correct!

A structure where any element can be removed in $O(1)$ time.

8. Which operation removes the element at the front of a queue?

peek

push

dequeue

✓ Correct!

pop

9. What is the typical average-case time complexity to look up a value by key in a hash map?

$O(n)$ because all keys must be scanned sequentially.

$O(\log n)$ due to binary search within buckets.

$O(1)$ on average with a good hash function and low load factor.

✓ Correct!

$O(n \log n)$ because keys are sorted during insertion.



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Elements are stored in sorted order by default.

Elements are indexed by their insertion position.

It stores only unique elements (no duplicates).

✓ Correct!

Duplicate values are allowed and kept together.

11. In a dynamic array, what is the worst-case time complexity of inserting an element at index i (not at the end)?

$O(1)$

$O(n)$

✓ Correct!

$O(\log n)$

$O(1)$ amortized

12. What is the time complexity of inserting a new node at the head of a singly linked list?

$O(n)$

$O(1)$

✓ Correct!

$O(\log n)$

$O(n \log n)$

13. Which operation returns the top element of a stack without removing it?



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✓ Correct!

✗ Incorrect.

14. Which of the following best describes a queue?

Last In, First Out (LIFO) with removals at the top.Elements are always kept in sorted order automatically.Random access to any index in $O(1)$ time.First In, First Out (FIFO) with `enqueue` at the back and `dequeue` at the front.

✓ Correct!

15. What is a hash collision in a hash map?

When the map runs out of memory and must be resized.When a key maps to multiple distinct values by design.When two different keys produce the same hash index.

✓ Correct!

When two identical keys are stored in different buckets.

16. Why do hash maps resize (rehash) as they grow?



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To compress values and reduce memory fragmentation.

To avoid triggering the language's garbage collector.

To keep the load factor low so that average operations remain $O(1)$.

✓ Correct!

17. Which statement about sets is true?

Membership tests are typically $O(1)$ on average.

✓ Correct!

Sets preserve insertion order by definition.

Set membership tests are $O(n \log n)$ on average.

Sets allow duplicate elements and keep counts.

18. Which time complexity grows faster than $O(n \log n)$ as n becomes large?

$O(1)$

$O(n^2)$

✓ Correct!

$O(\log n)$

$O(n)$

19. After implementing a brute-force solution, what is a good next step?

Discard tests and rewrite the solution from scratch.



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Analyze its time/space complexity and optimize identified bottlenecks.

✓ Correct!

Avoid considering edge cases to keep the code simple.

20. What does space complexity measure?

How many CPU cores a program uses.

How memory usage grows relative to input size.

✓ Correct!

The length of a program in lines of code.

How long a program takes to compile.

You have 19 out of 20 questions correct.

Submit and go to next challenge

Exit the quiz